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DSCI 550

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Homework 4

1. K-means

1. Final clustering result starting with A1 and A6 as centers:

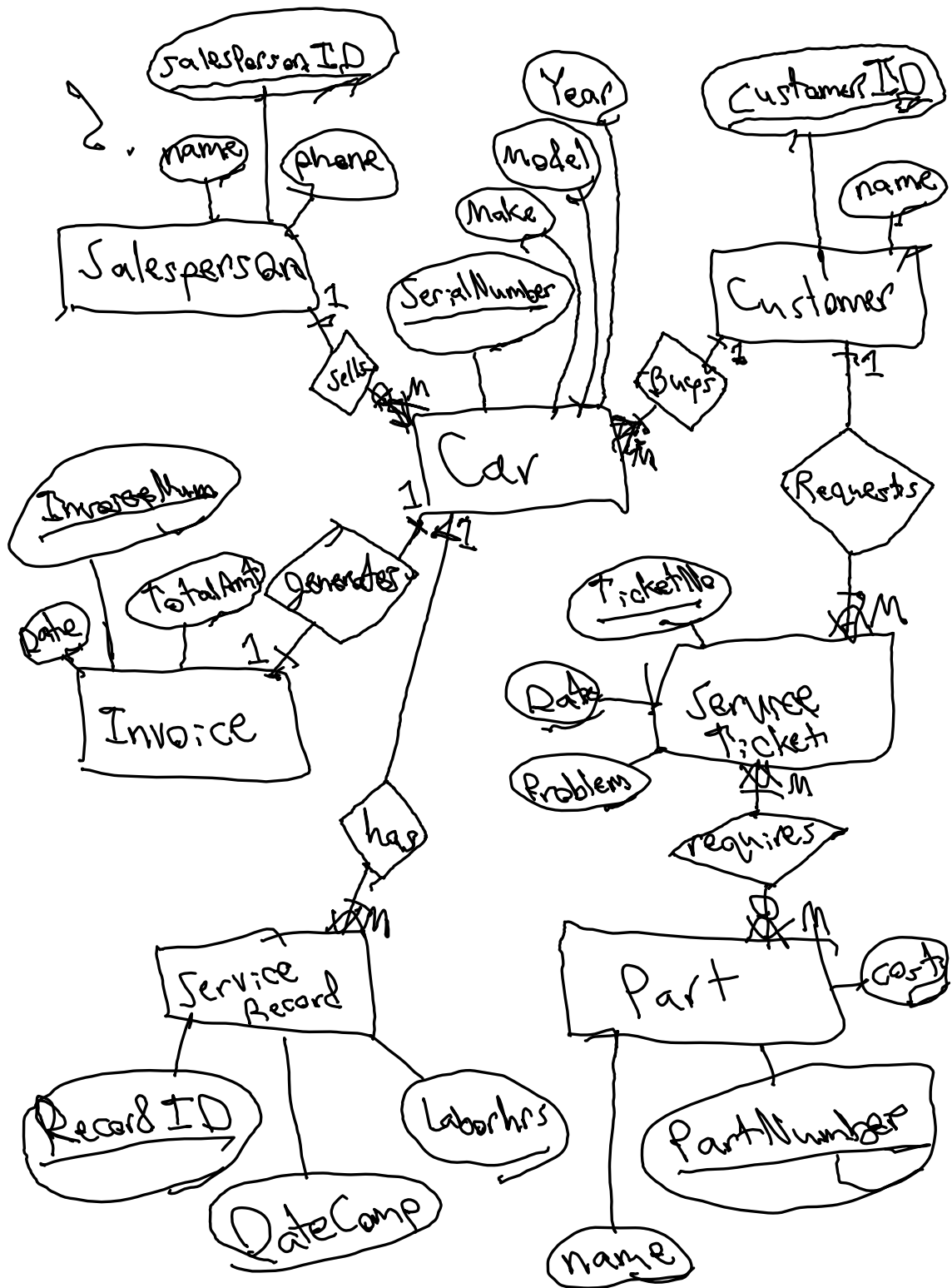
- i. **Initial Centers:** $C1 = A1(1, 2)$ and $C2 = A6(5, 4)$.
- ii. **Iteration 1:** We calculate the squared Euclidean distance from each point to $C1$ and $C2$.
 1. $A1(1,2)$: Closer to $C1$
 2. $A2(1,4)$: Closer to $C1$
 3. $A3(3,1)$: Closer to $C1$
 4. $A4(3,5)$: Closer to $C2$
 5. $A5(5,2)$: Closer to $C2$
 6. $A6(5,4)$: Closer to $C2$
 7. *Current Clusters:* Cluster 1 = $\{A1, A2, A3\}$, Cluster 2 = $\{A4, A5, A6\}$.
- iii. **New Centers:** We find the mean of the points in each cluster.
 1. $C1 \text{ (new)} = ((1+1+3)/3, (2+4+1)/3) = (1.67, 2.33)$
 2. $C2 \text{ (new)} = ((3+5+5)/3, (5+2+4)/3) = (4.33, 3.67)$
- iv. **Iteration 2:** Re-calculating distances to the new centers shows that the points are still closest to their current centers.
- v. **Final Clustering 1 Result:** * **Cluster 1:** $\{A1, A2, A3\}$
 1. **Cluster 2:** $\{A4, A5, A6\}$

2. Final clustering result starting with A3 and A4 as centers:

- i. **Initial Centers:** $C1 = A3(3, 1)$ and $C2 = A4(3, 5)$.
- ii. **Iteration 1:** * $A1(1,2)$: Closer to $C1$
 1. $A2(1,4)$: Closer to $C2$
 2. $A3(3,1)$: Closer to $C1$
 3. $A4(3,5)$: Closer to $C2$
 4. $A5(5,2)$: Closer to $C1$
 5. $A6(5,4)$: Closer to $C2$
 6. *Current Clusters:* Cluster 1 = $\{A1, A3, A5\}$, Cluster 2 = $\{A2, A4, A6\}$.
- iii. **New Centers:** * $C1 \text{ (new)} = ((1+3+5)/3, (2+1+2)/3) = (3, 1.67)$
 1. $C2 \text{ (new)} = ((1+3+5)/3, (4+5+4)/3) = (3, 4.33)$

- iv. **Iteration 2:** Re-calculating distances to the new centers confirms no points need to change clusters.
- v. **Final Clustering 2 Result:** * **Cluster 1:** {A1, A3, A5}
 - 1. **Cluster 2:** {A2, A4, A6}
- 3. Compute the quality using SSE and determine the better clustering:
 - i. The formula for SSE is: $SSE = \sum_{i=1}^k \sum_{x \in S_{\{i\}}} (x - \mu_{\{i\}})^2$
 - ii. **SSE for Clustering 1:**
 - 1. Cluster 1 SSE = $SSE_{\{cluster1\}} = (1 - 1.67)^2 + (2 - 2.33)^2 + (1 - 1.67)^2 + (4 - 2.33)^2 + (3 - 1.67)^2 + (1 - 2.33)^2 = 7.33$
 - 2. Cluster 2 SSE = $SSE_{\{cluster2\}} = (3 - 4.33)^2 + (5 - 3.67)^2 + (5 - 4.33)^2 + (2 - 3.67)^2 + (5 - 4.33)^2 + (4 - 3.67)^2 = 7.33$
 - 3. **Total SSE (Clustering 1) = $SSE_{\{Total_1\}} = 7.33 + 7.33 = 14.67$**
 - iii. **SSE for Clustering 2:**
 - 1. Cluster 1 SSE = $SSE_{\{cluster1\}} = (1 - 3)^2 + (2 - 1.67)^2 + (3 - 3)^2 + (1 - 1.67)^2 + (5 - 3)^2 + (2 - 1.67)^2 = 8.67$
 - 2. Cluster 2 SSE = $SSE_{\{cluster2\}} = (1 - 3)^2 + (4 - 4.33)^2 + (3 - 3)^2 + (5 - 4.33)^2 + (5 - 3)^2 + (4 - 4.33)^2 = 8.67$
 - 3. **Total SSE (Clustering 2) = $SSE_{\{Total_2\}} = 8.67 + 8.67 = 17.33$**
 - iv. **Conclusion:** Clustering 1 is better because a lower SSE indicates tighter, more cohesive clusters.

2. ER Diagram



3. Queries

1. SELECT * FROM Actor WHERE gender = 'M';
2. SELECT * FROM Actor WHERE fname = 'John' AND nationality = 'Spain';
3. SELECT fname, lname FROM Actor WHERE nationality = 'USA';
4. SELECT name, revenue FROM Movie WHERE studio = 'Universal Studio';
5. SELECT COUNT(*) FROM Movie WHERE year = 2018;
6. SELECT MAX(income) FROM Director;
7. SELECT name FROM Movie ORDER BY revenue DESC;
8. SELECT AVG(income) FROM Director WHERE gender = 'F';

4. Normalization

1. Is it in 1NF? Otherwise, make it in 1NF.
 - i. Yes, the table is already in First Normal Form (1NF). All attributes contain atomic (indivisible) values, and there are no repeating groups within a single row.
2. Based on your answer in 1), is it in 2NF? Otherwise, make it in 2NF.
 - i. No, it is not in 2NF. The primary key is the composite of (StudentID, CourseCode). However, there are *partial dependencies*—attributes that depend on only part of the primary key. For example, Student Name depends only on StudentID, and CourseName, Instructor, and InstructorPhone depend only on CourseCode.
 - ii. **To make it 2NF, decompose into three tables:**
 1. **Student** (<u>StudentID</u>, StudentName)
 2. **Course** (<u>CourseCode</u>, CourseName, Instructor, InstructorPhone)
 3. **Enrollment** (<u>StudentID</u>, <u>CourseCode</u>)
3. Based on your answer in 2), is it in 3NF? Otherwise, make it in 3NF.
 - i. **Answer:** No, it is not in 3NF. While it is in 2NF, the new Course table has a *transitive dependency*. The InstructorPhone depends on the Instructor, which in turn depends on the CourseCode.
 - ii. **To make it 3NF, decompose the Course table further:**
 1. **Student** (<u>StudentID</u>, StudentName)
 2. **Instructor** (<u>InstructorName</u>, InstructorPhone)
 3. **Course** (<u>CourseCode</u>, CourseName, InstructorName)
 4. **Enrollment** (<u>StudentID</u>, <u>CourseCode</u>)